

2. DATA TO INSIGHTS TO DECISIONS

DOU LI

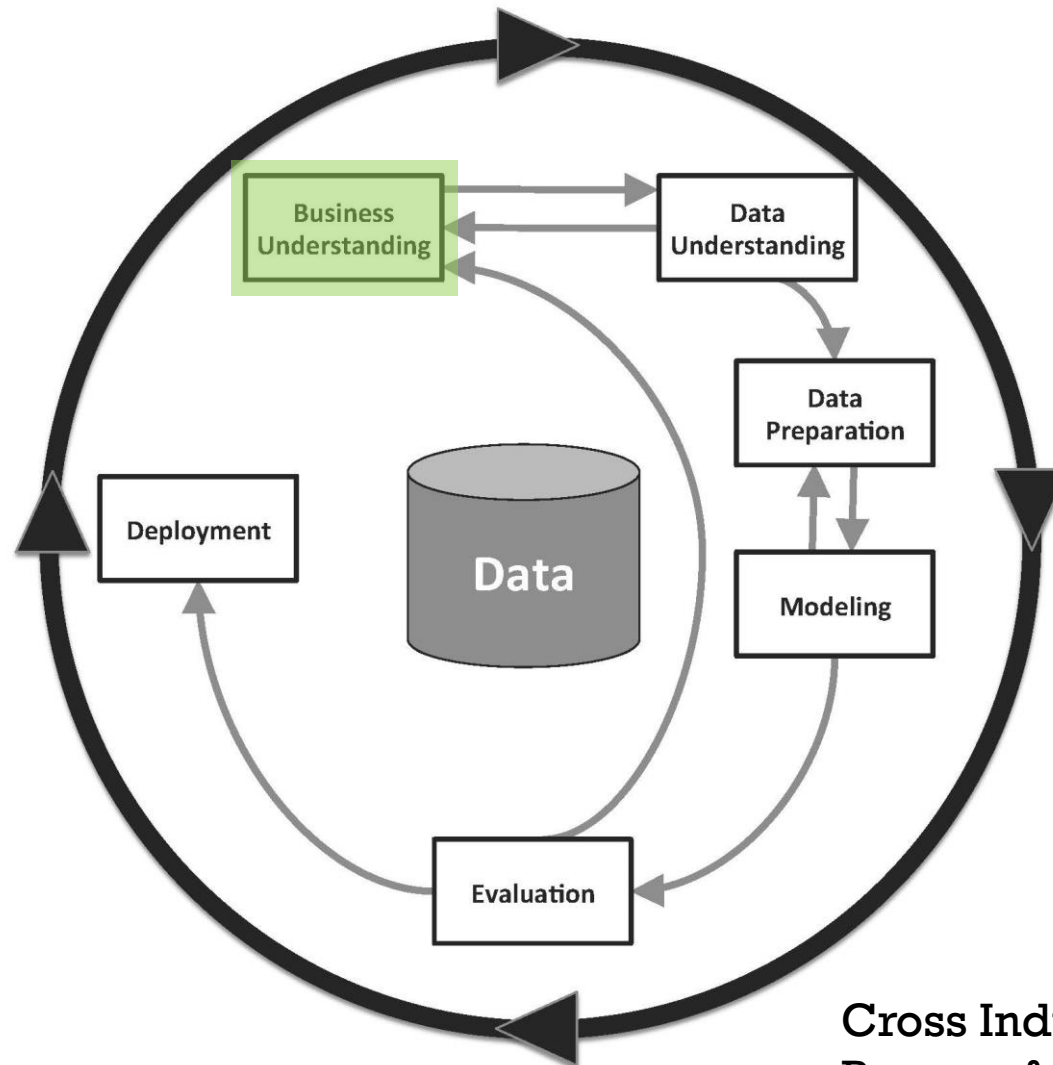
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Defining the analytics solution is the most important task in the **Business Understanding** phase of the CRISP-DM process.



Cross Industry Standard
Process for Data Mining

CONTENTS

- **Converting Business Problems into Analytics Solutions**
- **Assessing Feasibility**
- **Designing the Analytics Base Table**
- **Designing & Implementing Features**
- **Summary**

- **Converting a business problem into an analytics solution involves answering the following key questions:**
 - 1. What is the business problem?**
 - 2. What are the goals that the business wants to achieve?**
 - 3. How does the business currently work?**
 - 4. In what ways could a predictive analytics model help to address the business problem?**

CASE STUDY: MOTOR INSURANCE FRAUD

- In spite of having a fraud investigation team that investigates **up to 30%** of all claims made, a motor insurance company is still **losing too much money** due to fraudulent claims.
- What predictive analytics solutions could be proposed to help address this business problem?

CASE STUDY: MOTOR INSURANCE FRAUD

Brainstorming!

- Potential analytics solutions include:
 - ✓ Claim prediction
 - to predict the likelihood that an insurance claim is fraudulent.
 - ✓ Member prediction
 - to predict the propensity of a member to commit fraud in the near future.
 - ✓ Application prediction
 - to predict, at the point of application, the likelihood that a policy someone has applied for will ultimately result in a fraudulent claim.
 - ✓ Payment prediction
 - to predict the amount most likely to be paid out by an insurance company after having investigated a claim.

CONTENTS

- Converting Business Problems into Analytics Solutions
- **Assessing Feasibility**
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- Evaluating the feasibility of a proposed analytics solution involves considering the following questions:
 1. Is the **data** required by the solution available, or could it be made available?
 2. What is the **capacity** of the business to utilize the insights that the analytics solution will provide?

CASE STUDY: MOTOR INSURANCE FRAUD

- What are the **data** and **capacity** requirements for the proposed **Claim Prediction** analytics solution for the motor insurance fraud scenario?

CASE STUDY: MOTOR INSURANCE FRAUD

[Claim prediction]

- *Data Requirements:*

- A large collection of historical claims marked as 'fraudulent' and 'non-fraudulent'.
- The details of each claim, the related policy, and the related claimant would need to be available.

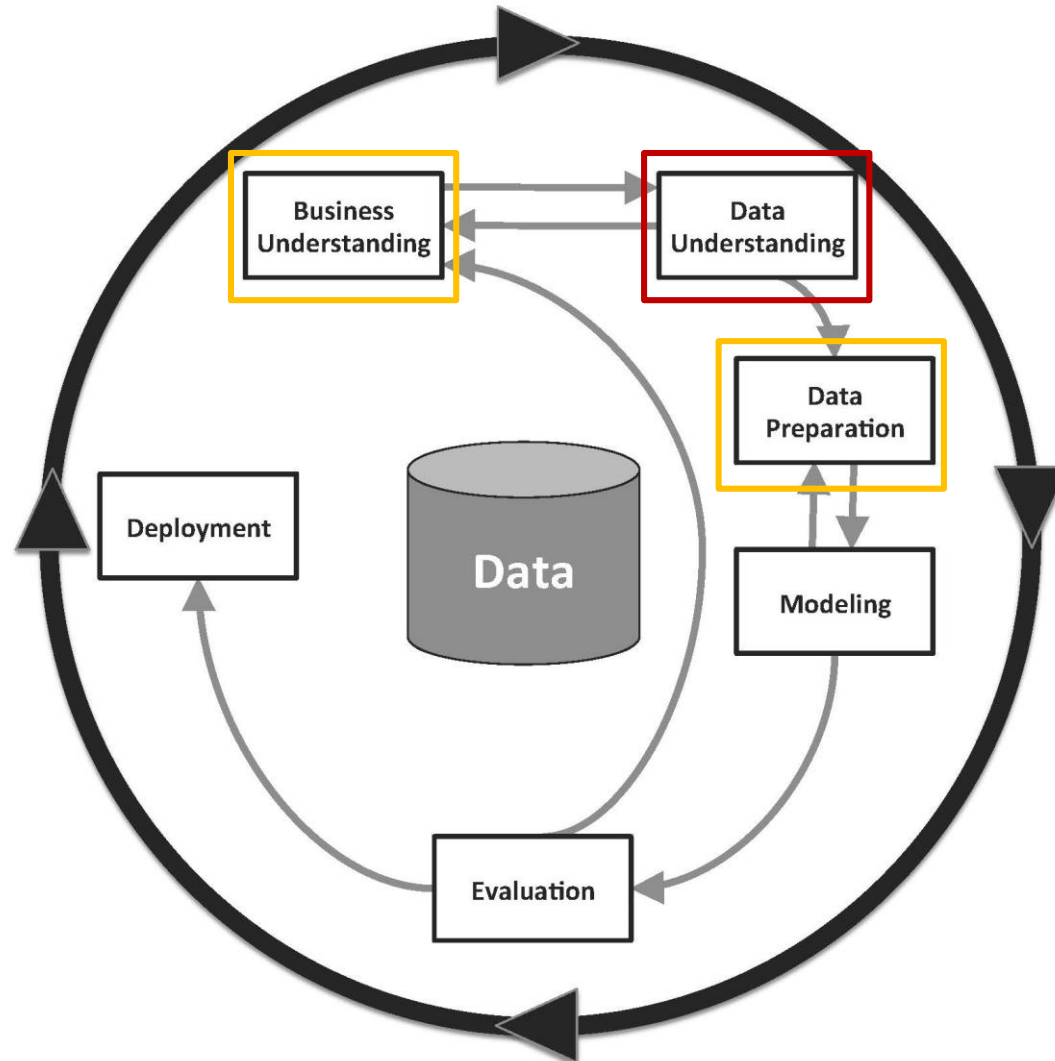
- *Capacity Requirements:*

- A mechanism could be put in place to inform claims investigators that some claims were prioritized above others.
- Information about claims become available in a suitably timely manner so that the claims investigation process would not be delayed by the model.

CONTENTS

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- **ABT design** sits primarily in the **Data Understanding** phase of the **CRISP-DM** process, but also overlaps with the **Business Understanding** and **Data Preparation** phases.



- The basic structure in which we capture historical datasets is the **Analytics Base Table (ABT)**

Descriptive Features							Target Feature
----	----	----	----	----	----	----	----
----	----	----	----	----	----	----	----
----	----	----	----	----	----	----	----
----	----	----	----	----	----	----	----
----	----	----	----	----	----	----	----

feature

instance

Table2.1: The general structure of an **analytics base table** — descriptive features and a target feature.

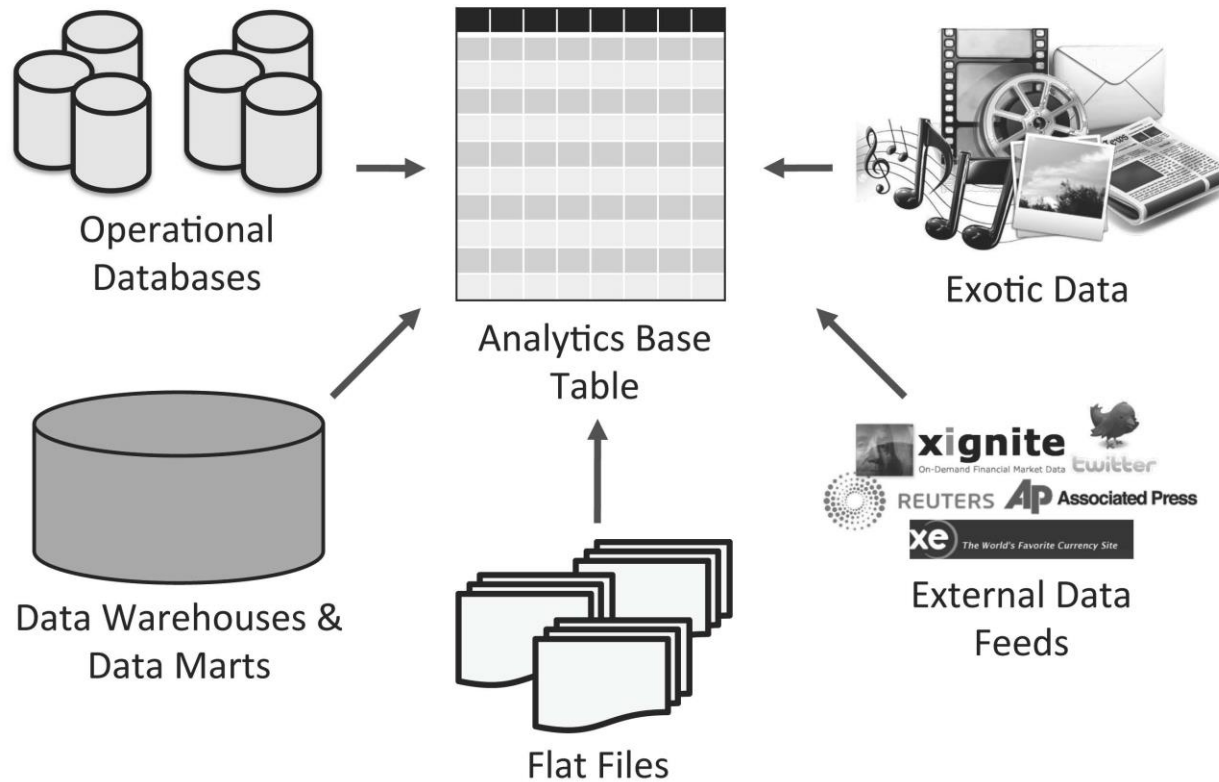


Figure2.1: The different data sources typically combined to create an analytics base table.

- The **prediction subject** defines the basic level at which predictions are made, and each row in the ABT will represent **one instance** of the prediction subject—the phrase **one-row-per-subject** is often used to describe this structure.
- Each row in an ABT is composed of a set of descriptive features and a target feature.
- **Defining features can be difficult!**
- A good way to define features is to identify the key **domain concepts** and then to base the features on these concepts.

- A **domain concept** is a **high-level abstraction** that describes some characteristic of the prediction subject from which we derive a set of **concrete features** that will be included in an ABT.

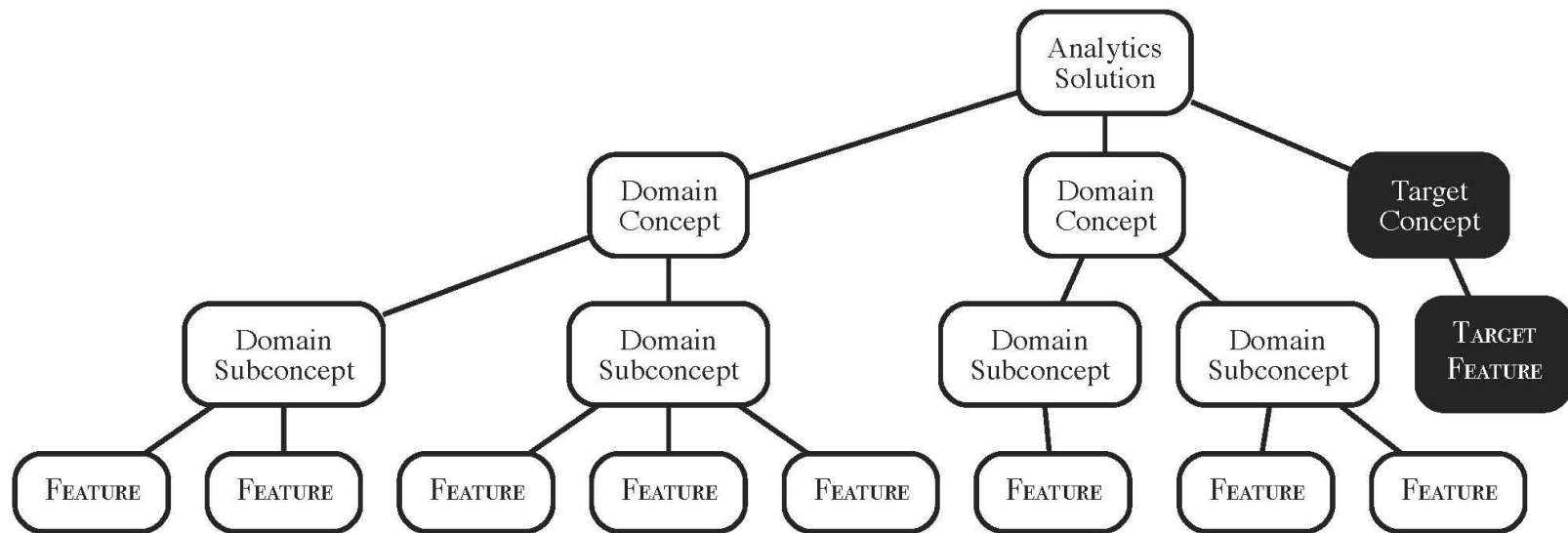


Figure 2.2: The [hierarchical relationship](#) between an analytics solution, domain concepts, and descriptive features.

- There are a number of general domain concepts that are often useful:
 - Prediction Subject Details
 - Demographics
 - E.g.: age, gender, occupation, address...
 - Usage
 - Changes in Usage
 - Special Usage
 - Lifecycle Phase
 - E. g.: a new customer, a loyal customer or a lapsing customer?
 - Network Links

CASE STUDY: MOTOR INSURANCE FRAUD

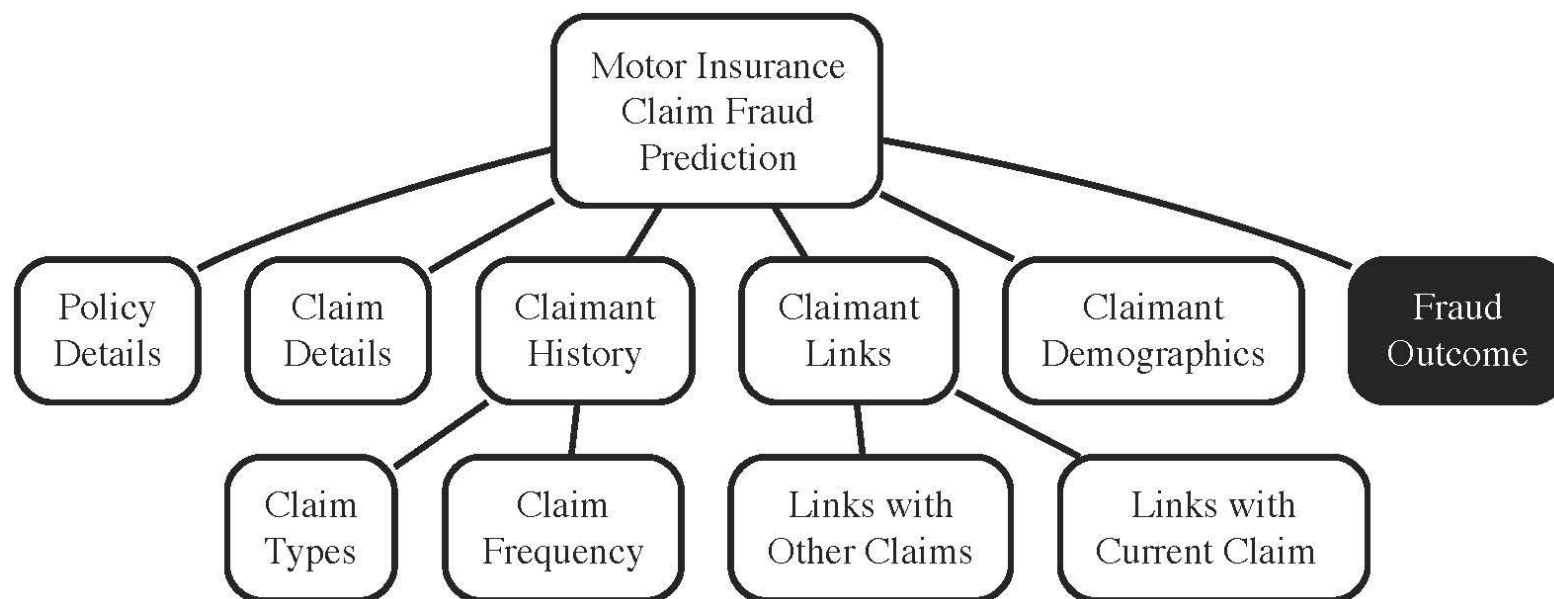


Figure 2.3: Example domain concepts for a motor insurance fraud claim prediction analytics solution.

CONTENTS

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- Three key data considerations are particularly important when we are designing features.
 - **Data availability**
 - **Timing**
 - **Longevity**

DIFFERENT TYPES OF DATA

continuous

- **Numeric:** True numeric values that allow arithmetic operations (e.g., price, age)
- **Interval:** Values that allow ordering and subtraction, but do not allow other arithmetic operations (e.g., date, time)

categorical

- **Ordinal:** Values that allow ordering but do not permit arithmetic (e.g., size measured as small, medium, or large)
- **Categorical:** A finite set of values that cannot be ordered and allow no arithmetic (e.g., country, product type)
- **Binary:** A set of just two values (e.g., gender)
- **Textual:** Free-form, usually short text data (e.g., name, address)

DIFFERENT TYPES OF DATA

ID	NAME	DATE OF BIRTH	GENDER	CREDIT RATING	COUNTRY	SALARY
0034	Brian	22/05/78	male	aa	ireland	67,000
0175	Mary	04/06/45	female	c	france	65,000
0456	Sinead	29/02/82	female	b	ireland	112,000
0687	Paul	11/11/67	male	a	usa	34,000
0982	Donald	01/12/75	male	b	australia	88,000
1103	Agnes	17/09/76	female	aa	sweden	154,000

Figure 2.4: Sample descriptive feature data illustrating numeric, binary, ordinal, interval, categorical, and textual types.

DIFFERENT TYPES OF FEATURES

- The features in an ABT can be of two types:
 - **raw features**
 - **derived features**
- There are a number of common derived feature types:
 - **Aggregates**
 - E.g., count, sum, average, minimum, or maximum of the values...
 - **Flags**
 - Binary features that indicate presence or absence of some characteristic within a dataset.
 - **Ratios**
 - Ratios are continuous features that capture the relationship between two or more raw data values.
 - **Mappings**
 - Mappings are used to convert continuous features into categorical features
 - often used to reduce the number of unique values that a model will have to deal with.

HANDLING TIME

- Many of the predictive models that we build are **propensity models**, which inherently have a temporal element
- For **propensity modeling**, there are two key periods:
 - **observation period**
 - **outcome period**

HANDLING TIME

- In some cases the observation and outcome period are measured over **the same time** for all predictive subjects.

The launch of new product: 华为Mate XT 2, IPHONE Duo

Will a customer buy it based on past shopping behavior?

2012							2013					
Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	
		Observation Period							Outcome Period			

(a) Observation period and outcome period

[illegible]

each line represents
a customer

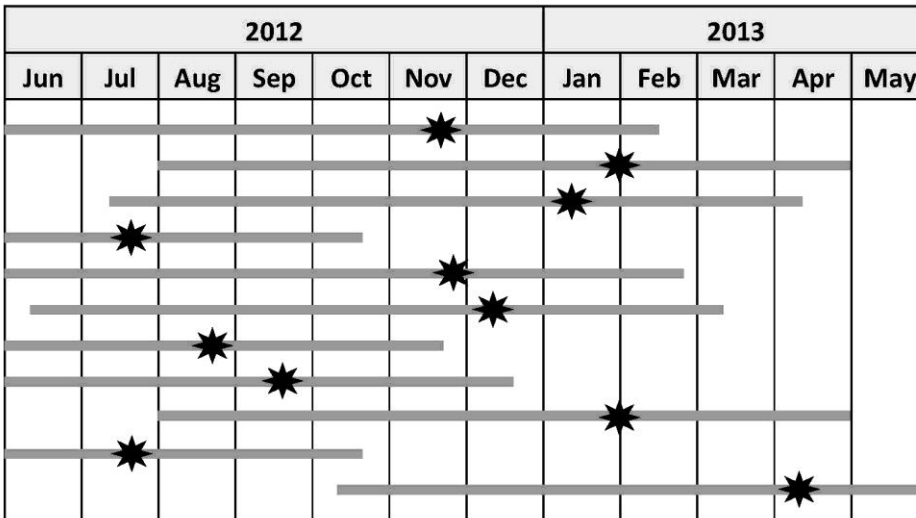
(b) Observation and outcome periods for multiple customers

Figure 2.5: Modeling points in time.

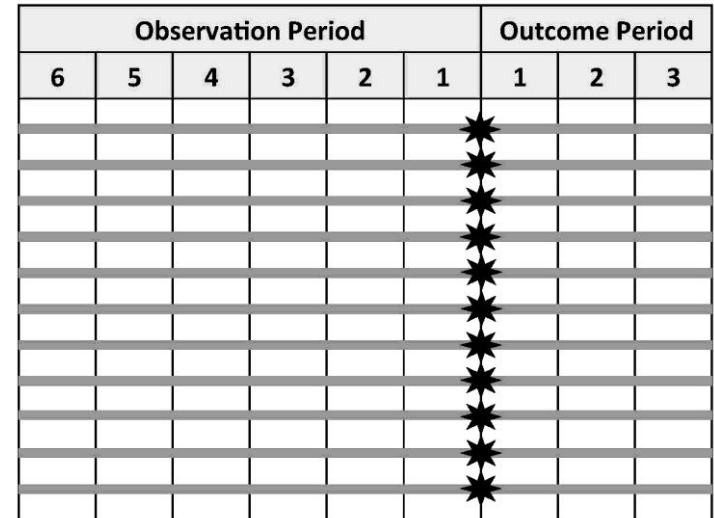
HANDLING TIME

- Often the observation period and outcome period will be measured over **different dates** for each prediction subject.

Insurance claims fraud scenario



(a) Actual

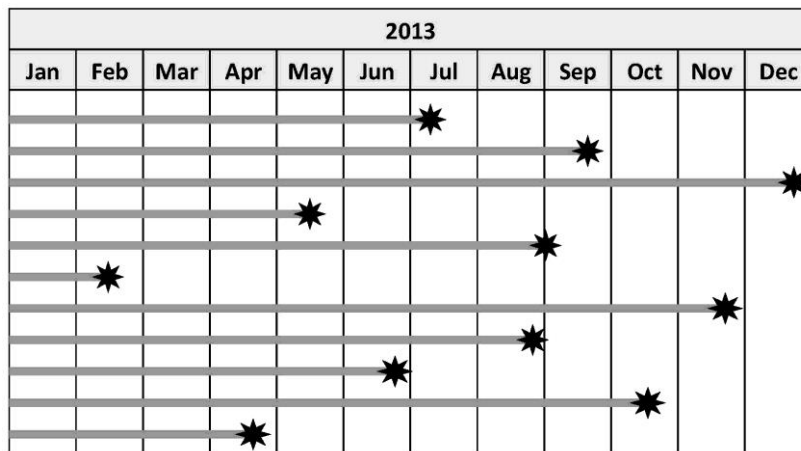


(b) Aligned

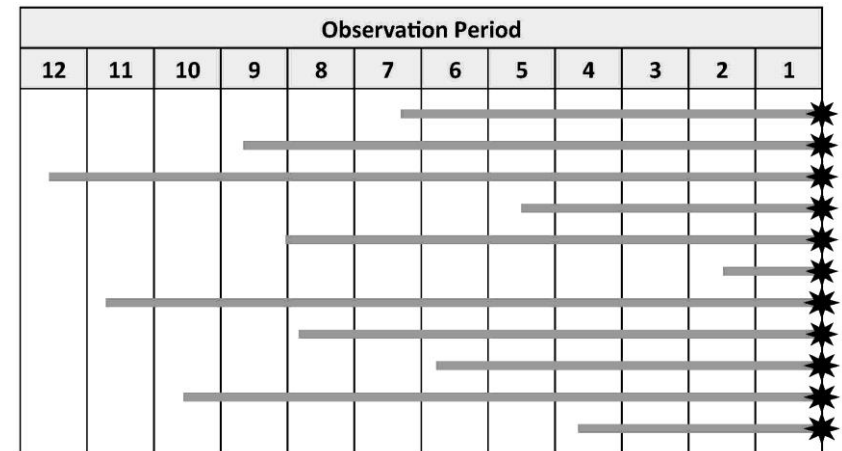
Figure 2.6: Observation and outcome periods defined by **an event** rather than by a fixed point in time (each line represents a prediction subject and stars signify events).

HANDLING TIME

- In some cases only the descriptive features have a time component to them, and **the target feature is time independent**.
- A **next-best-offer model** is used to determine the least expensive incentive that needs to be offered to a customer who is considering canceling a service, e. g., a mobile phone contract



(a) Actual

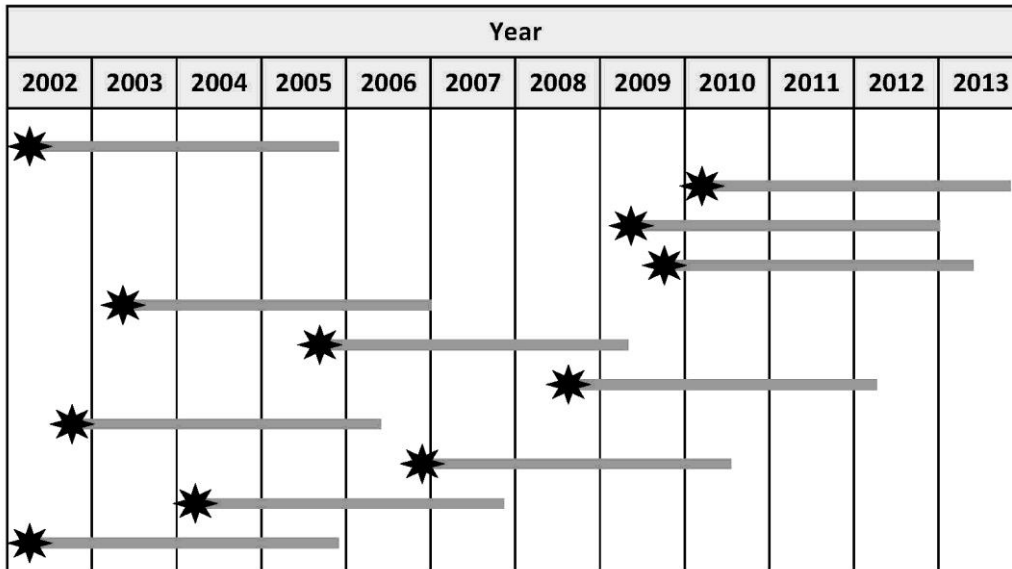


(b) Aligned

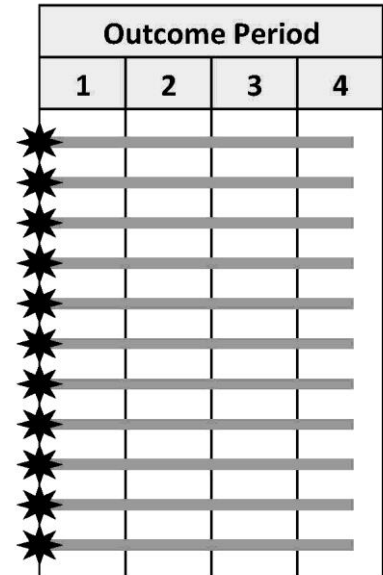
Figure 2.7: Modeling points in time for a scenario with **no real outcome period** (each line represents a customer, and stars signify events).

HANDLING TIME

- Conversely, the target feature may have a time component and **the descriptive features are time independent**.
- **Loan default prediction** is an example where the definition of the target feature has a time element but the descriptive features are time independent.



(a) Actual



(b) *Aligned*

Figure 2.8: Modeling points in time for a scenario with **no real observation period** (each line represents a customer, and stars signify events).

LEGAL ISSUES

- Data analytics practitioners can often be frustrated by **legislation** that stops them from including features that appear to be particularly well suited to an analytics solution in an ABT.
- There are significant differences in legislation in different **jurisdictions**, but a couple of key relevant principles almost always apply.
 - **Anti-discrimination legislation**
 - **Data protection legislation**

LEGAL ISSUES

- Although, data protection legislation changes significantly across different jurisdictions, there are some common **tenets** on which there is broad agreement which affect the design of ABTs
 - The **collection limitation principle**
 - The **purpose specification principle**
 - The **use limitation principle**

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新京报 | 热点 |

据网信中国官微消息，根据网络安全审查结论及发现的问题和线索，国家互联网信息办公室依法对滴滴全球股份有限公司涉嫌违法行为进行立案调查。经查实，滴滴全球股份有限公司违反《网络安全法》《数据安全法》《个人信息保护法》的违法违规行为事实清楚、证据确凿、情节严重、性质恶劣。

7月21日，国家互联网信息办公室依据《网络安全法》《数据安全法》《个人信息保护法》《行政处罚法》等法律法规，对滴滴全球股份有限公司处人民币80.26亿元罚款，对滴滴全球股份有限公司董事长兼CEO程维、总裁柳青各处人民币100万元罚款。

IMPLEMENTING FEATURES

- Implementing a **derived feature**, however, requires data from multiple sources to be combined into a set of single feature values.
- A few key **data manipulation operations** are frequently used to calculate derived feature values:
 - joining data sources
 - filtering rows in a data source
 - filtering fields in a data source
 - deriving new features by combining or transforming existing features
 - aggregating data sources

CASE STUDY: MOTOR INSURANCE FRAUD

- What are the **observation period** and **outcome period** for the motor insurance claim prediction scenario?
- The observation period and outcome period are measured over different dates for each insurance claim, defined relative to **the specific date of that claim**.
- The **observation period** is the time prior to the claim event, over which the descriptive features capturing the claimant's behavior are calculated.
- The **outcome period** is the time immediately after the claim event, during which it will emerge whether the claim is fraudulent or genuine.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Frequency** domain concept?

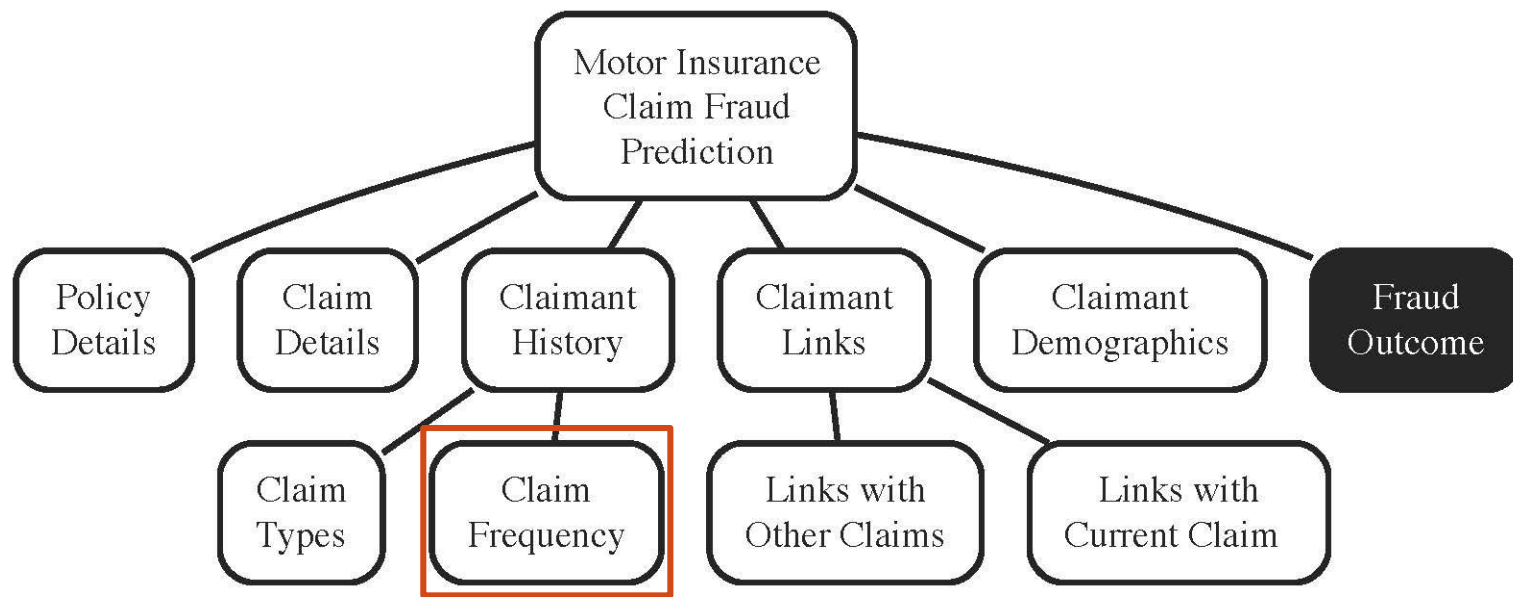


Figure 2.3: Example domain concepts for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Frequency** domain concept?

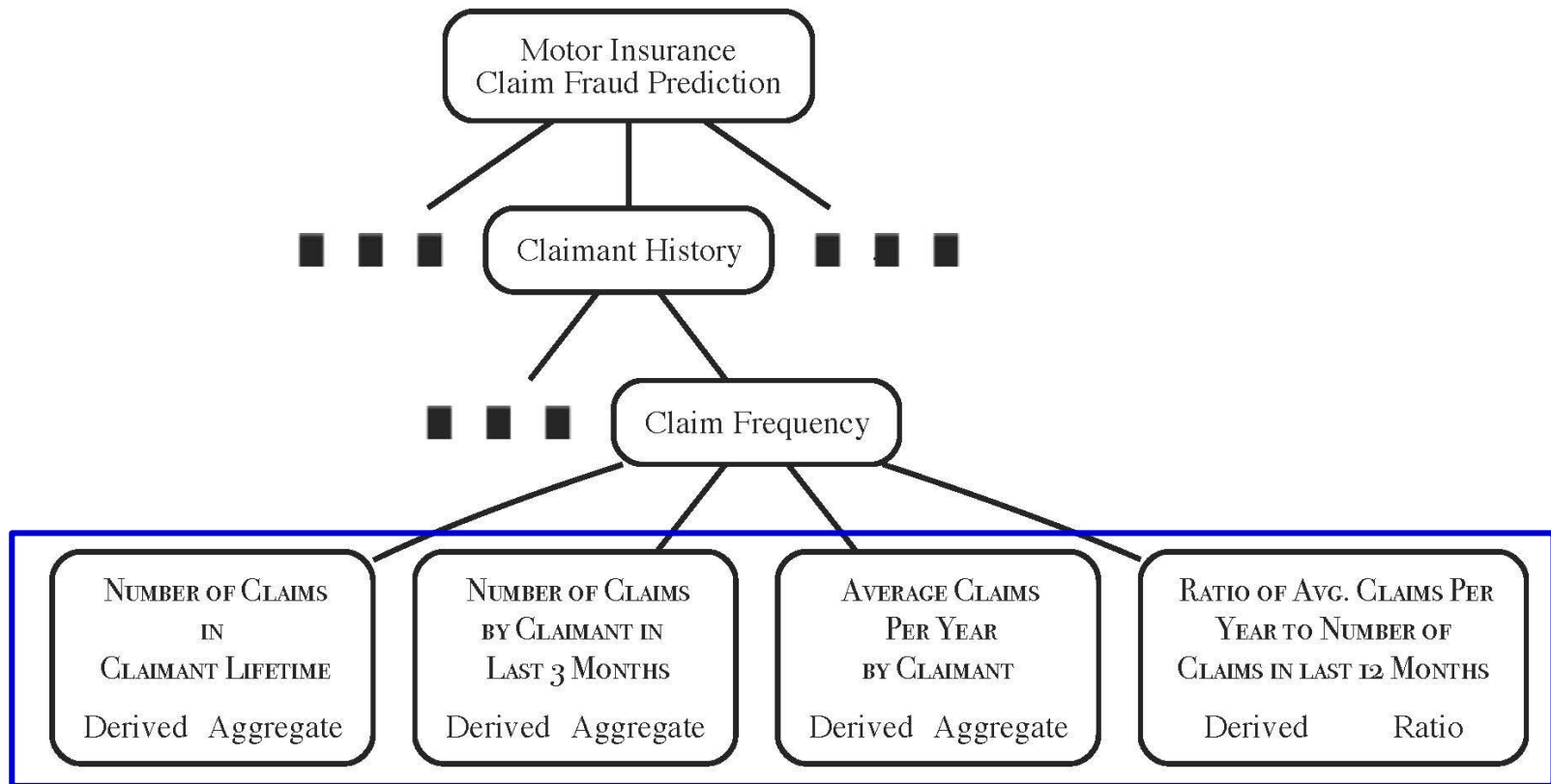


Figure 2.9: A subset of the domain concepts and related features for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Types** domain concept?

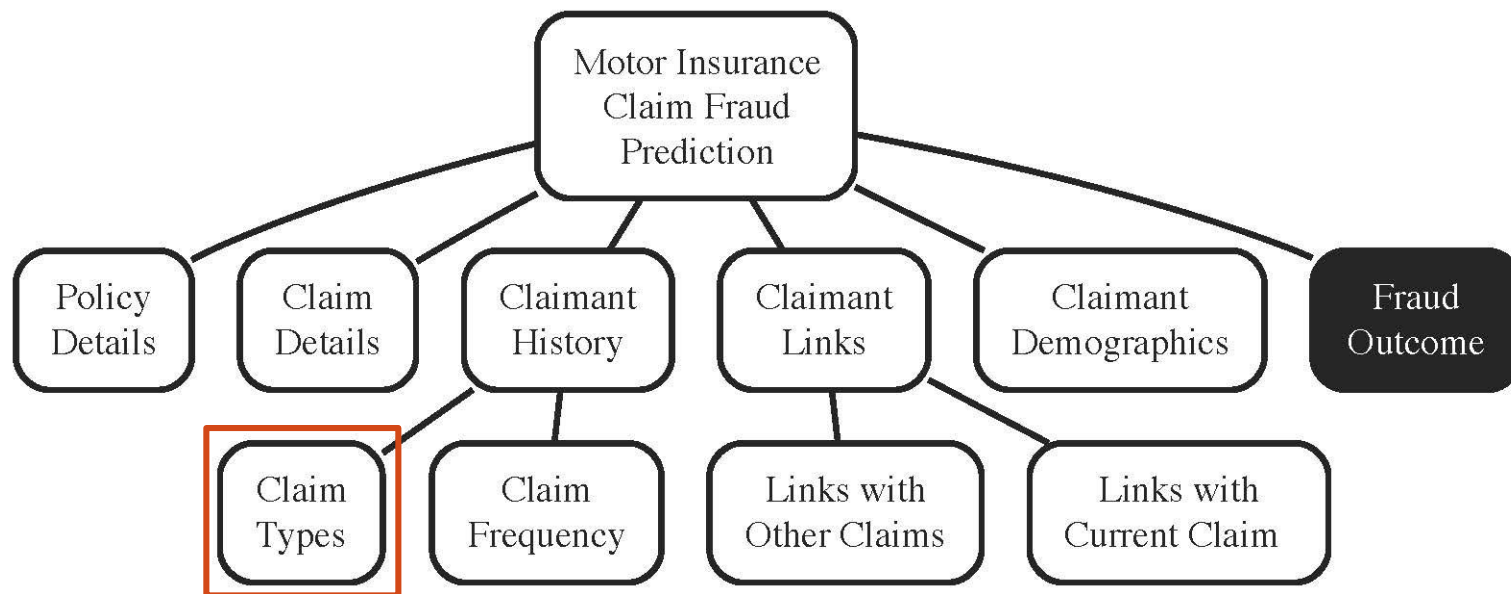


Figure 2.3: Example domain concepts for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Types** domain concept?

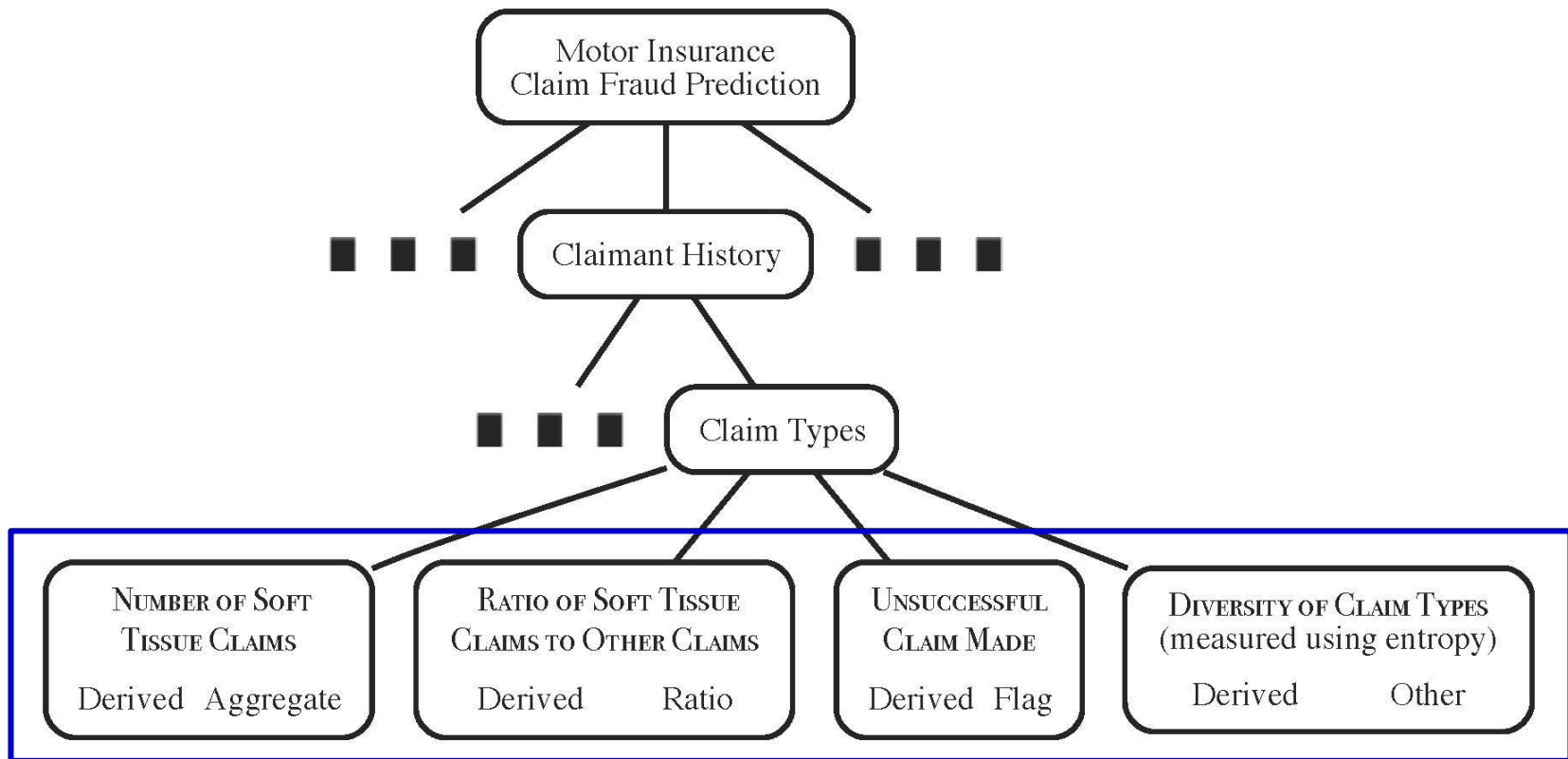


Figure 2.10: A subset of the domain concepts and related features for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Details** domain concept?

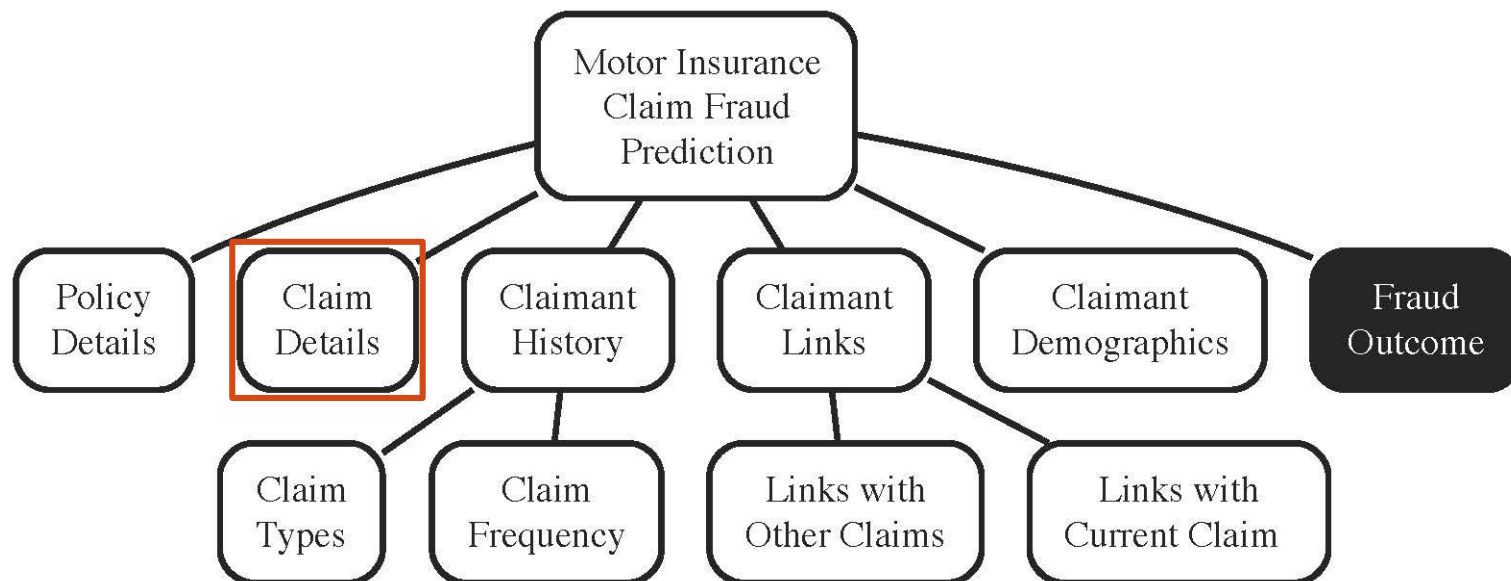


Figure 2.3: Example domain concepts for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- What features could you use to capture the **Claim Details** domain concept?

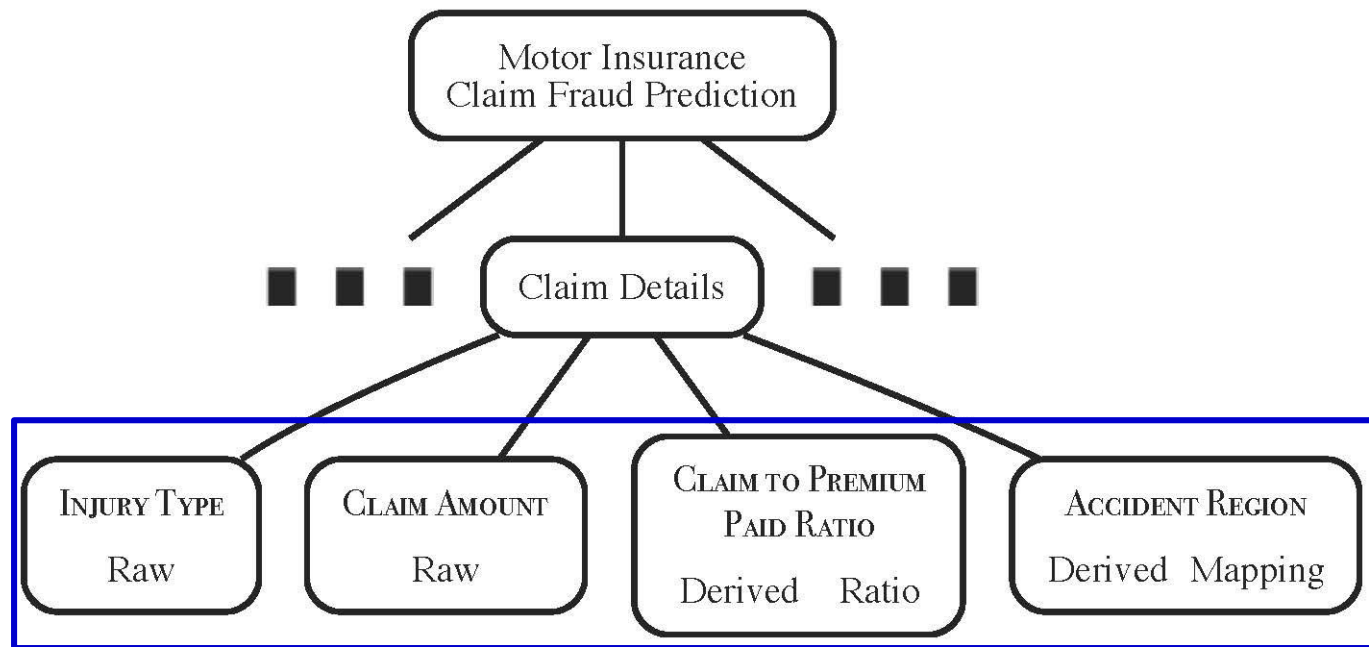


Figure 2.11: A subset of the domain concepts and related features for a motor insurance fraud prediction analytics solution.

CASE STUDY: MOTOR INSURANCE FRAUD

- The following table illustrates the structure of the final ABT that was designed for the motor insurance claims fraud detection solution.
- The table contains more descriptive features than the ones we have discussed
- The table also shows the first four instances.
- If we examine the table closely, we see a number of **strange values** (for example, - 9 999) and a number of **missing values**—we will return to these in Chapter 3.

- **Table 2.2:** The ABT for the motor insurance claims fraud detection solution.

ID	TYPE	INC.	MARITAL STATUS	NUM. CLMNTS.	INJURY TYPE	HOSPITAL STAY	CLAIM AMT.
1	CI	0	Married	2	Soft Tissue	No	1 625
2	CI	0		2	Back	Yes	15 028
3	CI	54 613		1	Broken Limb	No	-9 999
4	CI	0		3	Serious	Yes	270 200
			⋮			⋮	

ID	TOTAL CLAIMED	NUM. CLAIMS	NUM. CLAIMS 3 MONTHS	AVG. CLAIMS PER YEAR	AVG. CLAIMS RATIO	NUM. SOFT TISSUE	% SOFT TISSUE
1	3 250	2	0	1	1	2	1
2	60 112	1	0	1	1	0	0
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
		⋮			⋮		

ID	UNSUCC. CLAIMS	CLAIM AMT. REC.	CLAIM DIV.	CLAIM TO PREM.	REGION	FRAUD FLAG
1	2	0	0	32.5	MN	1
2	0	15 028	0	57.14	DL	0
3	0	572	0	-89.27	WAT	0
4	0	270 200	0	30.186	DL	0
		⋮			⋮	

- **Table 2.2:** The ABT for the motor insurance claims fraud detection solution.

NUM. CLMNTS.	NUMBER OF CLAIMANTS
NUM. CLAIMS	NUMBER OF CLAIMS IN CLAIMANT LIFETIME
NUM. CLAIMS 3 MONTHS	NUMBER OF CLAIMS BY CLAIMANT IN LAST 3 MONTHS
AVG. CLAIMS PER YEAR	AVERAGE CLAIMS PER YEAR BY CLAIMANT
AVG. CLAIMS RATIO	RATIO OF AVERAGE CLAIMS PER YEAR TO NUMBER OF CLAIMS IN LAST 12 MONTHS
NUM. SOFT TISSUE	NUMBER OF SOFT TISSUE CLAIMS
% SOFT TISSUE	RATIO OF SOFT TISSUE CLAIMS TO OTHER CLAIMS
UNSUCC. CLAIMS	UNSUCCESSFUL CLAIM MADE
CLAIM DIV.	DIVERSITY OF CLAIM TYPES
CLAIM AMT.	CLAIM AMOUNT
CLAIM TO PREM.	CLAIM TO PREMIUM PAID RATIO
REGION	ACCIDENT REGION

CONTENTS

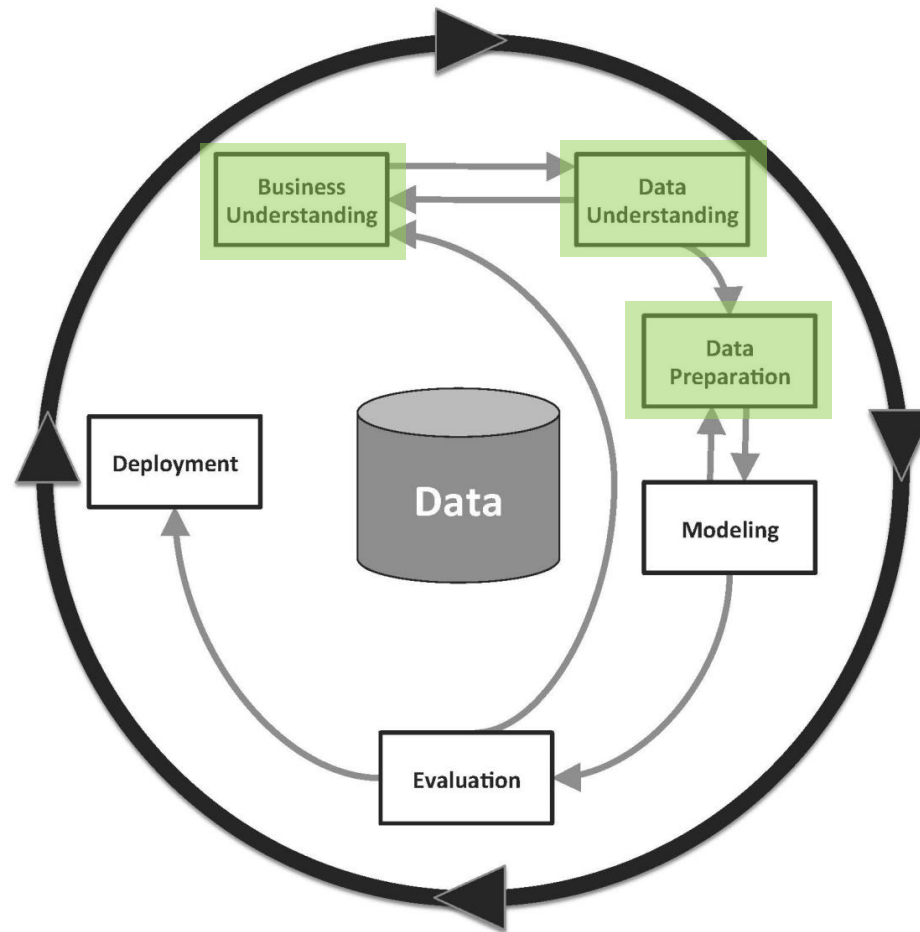
- Converting Business Problems into Analytics Solutions
- Assessing Feasibility
- Designing the Analytics Base Table
- Designing & Implementing Features
- **Summary**

- Predictive data analytics models built using machine learning techniques are tools that we can use to help **make better decisions** within an organization, not an end in themselves.
- It is important to fully understand the business problem that a model is being constructed to address—this is **the goal** behind converting business problems into analytics solutions

- Predictive data analytics models are reliant on the data that is used to build them—the **analytics base table (ABT)**.
- The first step in designing an ABT is to decide on the **prediction subject**.
- An effective way in which to design ABTs is to start by defining a set of **domain concepts** in collaboration with the business, and then designing **features** that express these concepts in order to form the actual ABT.

- Features (both descriptive and target) are concrete numeric or symbolic representations of domain concepts.
- It is useful to distinguish between **raw features** that come directly from existing data sources and **derived features** that are constructed by manipulating values from existing data sources.
- Common manipulations used in this process include aggregates, flags, ratios, and mappings, although any manipulation is valid.

- The techniques described here cover the **Business Understanding**, **Data Understanding**, and (partially) **Data Preparation** phases of the **CRISP-DM** process.



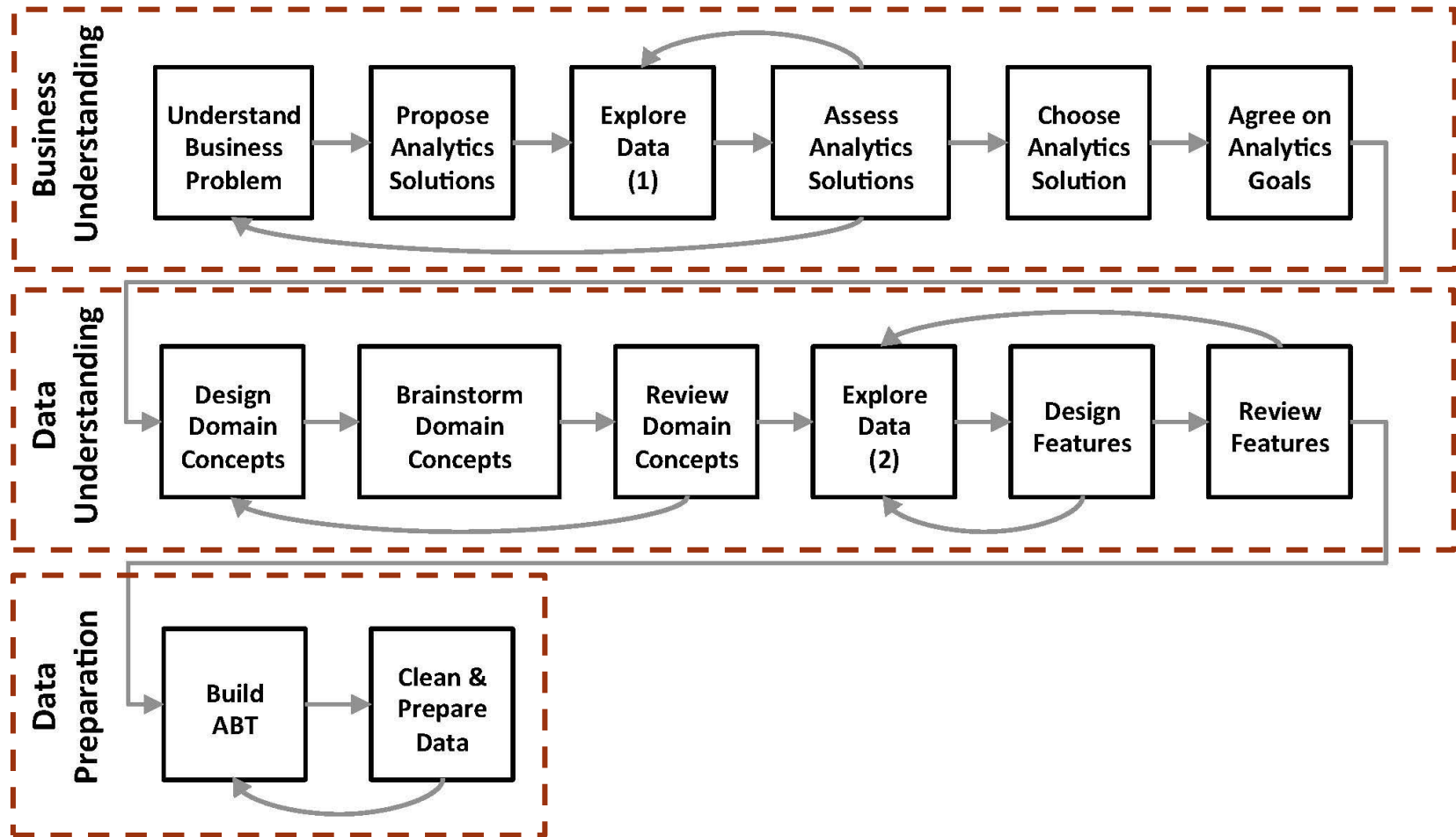


Figure 2.12: A summary of **the tasks** in the **Business Understanding**, **Data Understanding**, and **Data Preparation** phases of the **CRISP-DM** process.

HOMEWORK

- Reading: chapter 2
- Exercises: 2.7-3